Predicting outcomes in a football match using statistical techniques

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Abstract

We introduce a novel, unsupervised machine learning framework to find and validate team-specific playing structures for a football match. Using event data from the 2017/18 English Premier League season, we develop a hierarchical clustering algorithm that reduces the playing field from a 400-block spatial grid to a smaller number of distinct zones. The proposed framework is validated through both statistical analysis and domain-specific analysis. Quantitatively, the model demonstrates high stability and spatial cohesion, achieving an Adjusted Rand Index of 0.47 and a global Silhouette score of 0.30. Furthermore, the framework demonstrates high tactical accuracy, successfully differentiating between the possession-heavy positional play of Manchester City and the direct, long-ball transition model of Burnley.

1 Introduction

Quantifying performance in football is difficult as football is a continuous, low-scoring game where the most critical actions often occur away from the ball. For a long time, statistical analysis of football was limited to aggregate count metrics, such as number of successful passes, successful tackles, or possession percentage. While these metrics provide a high-level summary, they have a serious drawback – they ignore context. Traditional approaches rely on rigid, predefined pitch zones such as the breakdown of the pitch into thirds or fixed corridors. These frameworks fail to capture the dynamic nature of team formations and possession structures. Moreover, current football analytics remains heavily biased towards rare terminal events, with metrics like Expected Goals [2] valuing only shots while ignoring the regular actions such as passes and clearances.

In our report, we propose a framework to fill the gap between aggreagate metrics and tactical playing styles by quantifying football events like Pass, Shot, Clearance and all. Instead of quantifying metrics based on rigid, predefined pitch zones we try to cluster pitch zones based on playing

actions. To demonstrate the usefulness of this approach, we apply our framework to the 2017/18 Manchester City Centurions season — a dataset comprising over $28,000$ passes — due to their high-possession structure. Furthermore, we validate the robustness of our clusters using ARI and Silhouette Score by contrasting the results with teams with opposing tactical philosophies, such as the low-block systems of Burnley and West Bromwich Albion.

2 Literature Review

In 2011, [1] pointed out that not all passes are created equal. She applied markov chain to estimate expected value of goals in different states during football match. Later in 2012, Green [2] introduced the xG framework and popularized the specific term Expected Goal. This work used logistic regression to find the probability of a shot to turn into a goal. In recent years, more advanced frameworks like XGBoost algorithm (Anzer et al. [3]), Bayesian hierarchical model (Scholtes et al [4]) are also introduced. While the discussed approaches are highly suitable for modeling Expected Goals (xG), their application is fundamentally limited by the nature of the event being measured. A shot or a goal is a terminal event. Consequently, these models excel at estimating the probability. However, these frameworks are unsuitable for modeling non-terminal actions, such as passes or dribbles. A pass does not have an immediate binary success metric in terms of scoring; its value is derived from how it transitions the game from a low-threat state to a higher-threat state.

To create an impactful threat metric for football events, discretizing football pitch into functional tactical shapes is necessary. Rathke [5] showed that we can capture spatial context which significantly improves xG metrics by defining specific tactical zones. Singh [6] in his seminal work, showed that the pitch needs to be segmented into a grid to calculate transition probabilities. Another notable work in segmenting the pitch is given by Gyarmati and Stanojevic [7], who utilize a Markov chain-inspired methodology that partitions the pitch into team-specific clusters and computes the change in field value resulting from each pass. High dimensionality of the state space (1,000 clusters) relative to the sample size introduces significant sparsity in the transition matrix, potentially leading to high variance in field value estimation and a lack of robustness in player merit rankings. On other hand Singh [6]'s xT framework considers a rigid 16×12 grid structure. By implicitly assuming that all teams use the pitch in uniform manner this approach fails to capture playing tactics of specific teams. Moreover, both the framework only uses pass for valuation ignoring other football events.

Among these existing frameworks, we aim to build our threat metric based on the xT framework due to its high interpretability. However, as far as we know, no research has yet fixed two major issues: first, the grid used to divide the pitch is 'one-size-fits-all' and doesn't change based on a team's unique tactics. Second, the way outcomes are valued is too simple—it usually gives a 1 for a goal and a 0 for a miss, which ignores the value of a great chance that just happened to be missed.

We also noticed that most existing works focus only on the number of passes, leaving out other important events and failing to correctly value how different actions contribute to a team's overall threat.

3 xT Framework and Problem Statement

The Expected Threat (xT) framework proposed by Singh [6] utilizes a spatial discretization of the pitch into a 16×12 grid, resulting in 192 discrete cells to facilitate the calculation of transition probabilities. For every cell, four differrent probabilities are calculated.

- $m_{x,y}$ is the probability of moving the ball from cell (x, y)
- $s_{x,y}$ is the probability of shooting the ball from cell (x, y)
- $g_{x,y}$ is the probability that the short turns into a goal
- $T_{(x,y)\rightarrow(u,v)}$ is the transition probability or moving probability from cell (x, y) to cell (u, v)

Now suppose the expected threat or expected value for cell (x, y) is denoted by $V_{x,y}$. Given we shoot from cell (x, y) the expected payoff or expected value is $g_{x,y}$. Given we move from cell (x, y) expected payoff or expected value is $\Sigma_{(u,v)} T_{(x,y)\rightarrow(u,v)} V_{u,v}$. Thus the total expected payoff for cell (x, y) is given by

$$V_{x,y} = s_{x,y} \times g_{x,y} + m_{x,y} \times \sum_{(u,v)} T_{(x,y)\rightarrow(u,v)} V_{u,v}$$

To estimate value of $V_{x,y}$ they iteratively update value of $V_{x,y}$ until convergence with initial value as 0 for all $V_{x,y}$.

This above stated xT framework is an over-simplified version of markov decision process [9]. In general, a markov decision process is denoted by 4-tuple (S, A, P, R).

- S is the set of states or state space. Here set of all cells of the pitch is S.
- A is the set of actions at any state
- P is the transition probability i.e. $P_a(S, S')$ is the probability of transition to state S' from state S through action a
- R is the reward i.e. $R_a(S, S')$ is the reward for transition to state S' from state S through action a

In MDP, our objective is to find a policy π which is strategy for taking actions. Policy π is best if it maximize some pre-defined objective function. Here they consider expected discount sum over infinite horizon as the objective function, which is defined as

$$V^{\pi}(s) = E_{\pi}[\sum_{t=0}^{\infty} \gamma^t R_{a_t}(s_t, s_{t+1}) | s_t = s], \quad s \in S$$

where $\gamma \in [0, 1]$ is the discounting parameter. This can be simplified to

$$V^{\pi}(s) = \sum_{a \in A} \pi(a|s) \sum_{s' \in S} P_a(s, s')[R_a(s, s') + \gamma V^{\pi}(s')].$$

Now suppose that each state $s = (x, y)$ permits two possible actions: moving or shooting. We define the expected reward as $g_{x,y}$ if a shot is attempted, and zero otherwise. Then we can write

$$\begin{aligned}
V^{\pi}(s) &= \pi(move|s) \sum_{s' \in S} P_{move}(s, s')[R_{move}(s, s') + \gamma V^{\pi}(s')] + \\
&\pi(shoot|s) \sum_{s' \in S} P_{shoot}(s, s')[R_{shoot}(s, s') + \gamma V^{\pi}(s')] \\
&= \pi(move|s) \sum_{s' \in S} P_{move}(s, s') \gamma V^{\pi}(s') + \pi(shoot|s) \sum_{s' \in S} P_{shoot}(s, s') R_{shoot}(s, s') \\
&= m_{x,y} \times \gamma \times \sum_{(u,v)} T_{(x,y) \to (u,v)} V_{u,v} + s_{x,y} \times g_{x,y}
\end{aligned}$$

$V^{\pi}(s')$ is 0 when a shot is taken as shooting is the terminal event of any possession chain. Similarly $P_{shoot}(s, s')$ is always 1 and it terminates the chain of event. If we take discount factor γ to be 1, then we get back Singh's xT recursive equation.

While the xT framework provides a robust foundation for action valuation, it is characterized by several inherent methodological limitations.

- Creating metrics based on fixed grid cells assumes a uniform threat surface, ignoring the tactical differences across different teams.
- Use of empirical ratios to estimate transition probabilities leads to high variance in the presence of data sparsity.
- There is a lack of rigorous justification for the use of an infinite horizon ($\gamma = 1$) reward function. In a finite match duration, the assumption of an undiscounted cumulative reward requires more statistical validation.
- Training the model on a pooled dataset from multiple teams assumes a universal transition probability matrix, thereby neglecting the inherent heterogeneity in team-specific styles of play.

The primary objective of this work is to fill the methodological gap between rigorous statistical estimation and the Expected Threat (xT) framework. This study, therefore, seeks to reformulate the derivation of $V_{x,y}$ to ensure that the resulting threat surfaces are statistically significant, computationally stable, and tactically representative of diverse playing styles.

4 Methodology

To analyze the spatial zones in football, we analyze event data from the English Premier League 2017/2018 season, sourced from public dataset by Pappalardo [8]. We selected this season as it marks Manchester City's Centurions season, a benchmark for structured positional play. The continuous football pitch is first discretized into a regular grid of 20×20 blocks by mapping the Wyscout coordinates ($x, y \in [0, 100]$) to discrete indices. For each block b_i, we construct several role based features. These features, detailed in Table 1, are specifically selected to capture distinct distributions of offensive, defensive, and possession-oriented behaviors within the 400 blocks. The feature vectors are normalized using maximum scaling to rescale each attribute to the $[0, 1]$ range, thereby preventing volume bias for high volume events like Pass.

Table 1: Description of role-based features extracted for spatial clustering.

Role based feature	**Tactical Definition**
Fortress	Clearance, interception or defensive duels in the defensive half
Launch	Long forward pass in the defensive half
Entry	Short forward pass in the attacking half
Pivot	Lateral outward pass
Support	Lateral inward pass
Switch	Lateral long pass
Link	Lateral short pass
Safety	Backward pass
Isolation	Attacking duel or cross
Finish	Shot

To diminish boundary effects inherent in discretization and to incorporate information from local neighbouring blocks, we apply smoothing convolution using a custom 5×5 kernel K. This kernel assign a weight of 1 to the central block, 1/2 to first-order neighbors, and 1/4 to second-order neighbors, ensuring that each block's representation is informed by its immediate surroundings.

To measure the distance between blocks, we construct a hybrid distance matrix D_{hybrid}. The dissimilarity based on event occurrence between two blocks is measured using the Hellinger distance between the density of role based feature of each block. For the flow component, we compute the average pass direction vector $\vec{v}_i$ for each block. To minimize noise, these vectors are spatially smoothed using a 3×3 weighted kernel. We then calculate the Cosine Similarity S_{ij} between these

vectors and convert it to a bounded distance metric:

$$D_{\text{Flow}}(i,j) = \begin{cases} \frac{1-S_{ij}}{2} & \text{if } \vec{v}_i, \vec{v}_j \neq \vec{0} \\ 0.5 & \text{otherwise} \end{cases} \tag{1}$$

This transformation ensures that blocks with opposing flow will have distance 1, while blocks with no distinguishable flow are treated as neutral (0.5). The distance matrix of role based feature $D_{\text{Hellinger}}$ and the distance matrix based on flow direction D_{Flow} combined via a convex combination to set $D_{\text{hybrid}} = \alpha \cdot D_{\text{Hellinger}} + (1-\alpha) \cdot D_{\text{Flow}}$, with α set to 2/3 to prioritize role definition while retaining directional context. Using this precomputed distance matrix, we perform Agglomerative Hierarchical Clustering with the Complete Linkage to encourage compact, distinct clusters, applying a distance threshold of $\tau = 0.2$ to define the final functional zones.

The choice of α is chosen as 2/3 is also validated by a grid search between $[0.3, 0.7]$ with highest silhouette score. We discard value of α outside this interval as otherwise one matrix will solely determine clusters which is not our goal. For almost all the teams, $\tau = 0.2$ works better by keeping number of clusters in a desirable range (35+ to 45+) and also capturing underlying playing tactics.

We first assess the consistency of the clustering algorithm using a Split-Half test. The full season dataset ($N = 38$ matches) is randomly partitioned into two disjoint subsets, each containing 19 matches. The hierarchical clustering algorithm is applied independently to both subsets to generate two distinct cluster assignments. To measure the similarity between these independent clusterings, we calculate the Adjusted Rand Index (ARI) [10]. A high ARI indicates that the functional zones are stable tactical structures that are same across different subsets of matches.

Additionally, to evaluate the internal quality of the clusters on the full dataset, we compute the Silhouette Score [11]. We use a Label Permutation Test to verify that our tactical clusters are statistically cohesive and not arbitrary. By randomly shuffling cluster labels 1,000 times and comparing the resulting Silhouette Scores to our observed data, we set a p-value to 0.01 that confirms our clusters represent statistically significant, non-random functional zones of the pitch.

To validate the functional connectivity of the zones, we model the pitch as a directed weighted graph $G = (V, E)$. The nodes V represent the identified clusters, and the edges E represent successful passing transitions between them. The weight w_{ij} of an edge is defined as the frequency of passes from Cluster i to Cluster j. We define the distance between zones $d(i,j)$ as the inverse of this frequency w_{ij}. Using this network topology, we applied Dijkstra's Shortest Path Algorithm to identify optimal transition routes between defensive and offensive zones. We expect that if the clusters are capturing the true playing styles, then the shortest paths would align with known delivery mechanisms (e.g., cutbacks,through-balls). We further calculated Betweenness Centrality to identify clusters that serve as critical junctions in possession sequences.

Finally, to verify that the discrete clusters retain sufficient spatial resolution to distinguish individual player roles, we perform a test by constructing heatmaps from our findings. For a given player P, we compute a Vector $\mathbf{v}_P \in \mathbb{R}^K$, where each entry represents the percentage of events exhibited by player P in every clusters.

We then generate a Heatmap $H \in \mathbb{R}^{20\times 20}$ by mapping these percentage back onto the spatial grid. For every block b_{ij}, the heat value was assigned based strictly on its cluster membership: $H_{ij} = v_{P,L(b_{ij})}$. This reconstructed map is visually compared to the ground-truth Kernel Density Estimate (KDE) of the player's raw event coordinates. High similarity between the reconstructed map and the ground truth confirms that the clustering algorithm successfully captures the variance in player positioning (e.g., distinguishing a "Box Striker" from a "False 9") despite the dimensionality reduction.

5 Results

To explain our next steps for our tactical analysis of specific playing patterns, we examine our clusters for Manchester City (Team 1625) by applying our clustering algorithm. Figure 1 displays the final 39 discretized functional clusters by reducing dimensions from a 400-block grid.

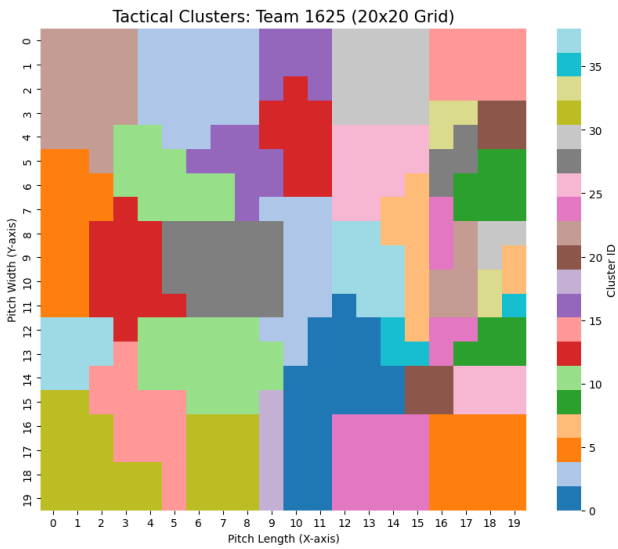


Figure 1: 39 clusters of Manchester City by applying our algorithm

Though our algorithm clusters blocks based on only role based features, the resulting clusters form a contiguous shapes. It confirms that neighbouring areas share tactical patterns. Furthermore these clusters do not form any symmetrical shapes, correctly identifying the asymmetrical playing style of Guardiola's side. In essence, by squeezing the pitch into these 39 functional zones, this model

strictly reduces spatial noise by only keeping meaningful playing zones. In subsequent analysis we validate our model both statistically as much as based on our domain knowledge.

5.1 Model Validation

To confirm that our clusters represent underlying tactical zones rather than random partitions, we conduct a label permutation test. Figure 2 illustrates the null distribution of silhouette score for two teams: league champion Manchester City and relegated side West Bromwich. We randomly permute clustering labels 1000 times for this test. For both these teams, the actual silhouette score is far greater than the null scores obtained under random partitions. This test confirms that our clustering algorithm captures statistically significant zones based on a team's playing patterns.

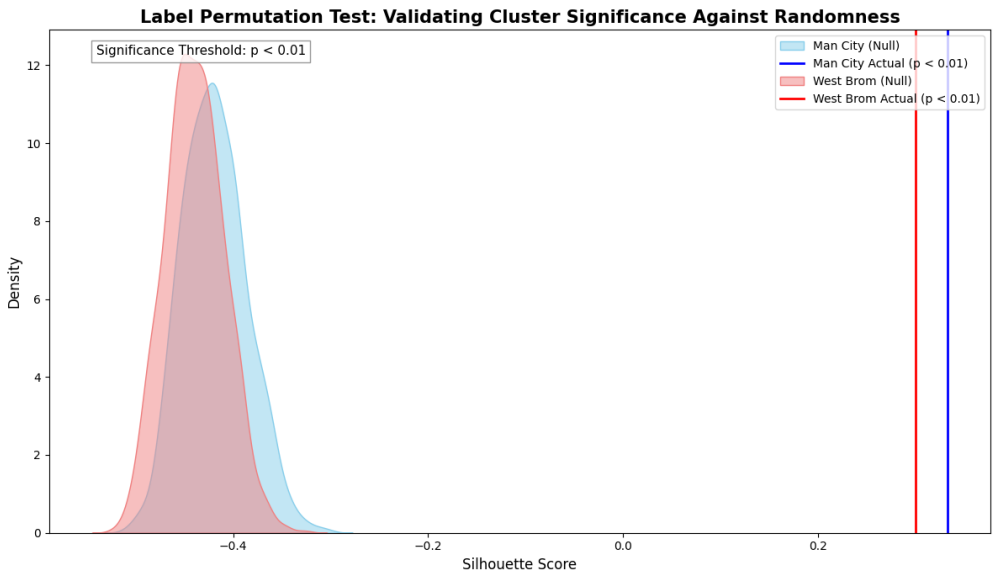


Figure 2: Results from the permutation test with 1000 permutation

To validate our clustering algorithm gives stable clusters, we have partitioned specific teams matches in two parts and then calculated ARI for cluster of both parts. From table 2 league's top ranked teams like Manchester cIty, Manchester United, Liverpool show high ARI metric ($ARI \approx 0.47$–0.48). For all 20 teams the ARI values range from 0.4 to 0.5. This indicates that our algorithm produces stable clusters regardless of a team's overall rankings.

Table 2: Comparison of cluster stability across all 20 teams (Organised based on league-standings)

Team	**ARI**	**Team**	**ARI**	**Team**	**ARI**	**Team**	**ARI**
Manchester City	0.47	Manchester United	0.48	Tottenham	0.48	Liverpool	0.48
Chelsea	0.51	Arsenal	0.51	Burnley	0.42	Everton	0.47
Leicester City	0.43	Newcastle United	0.48	Crystal Palace	0.45	Bournemouth	0.51
West Ham	0.43	Watford	0.46	Brighton	0.47	Huddersfield	0.52
Southampton	0.45	Swansea City	0.44	Stoke City	0.44	West Bromwich	0.41

In figure 3, Manchester City (Team 1625) exhibits highest silhouette score indicating their highly structured possession based playing style. Though the plot shows a stable silhouette score for all the teams around 0.3, for lower ranked team's silhouette score is little lesser that higher ranked teams. It suggests that lower ranked teams lack positional integrity as they play with defensive tactics mostly.

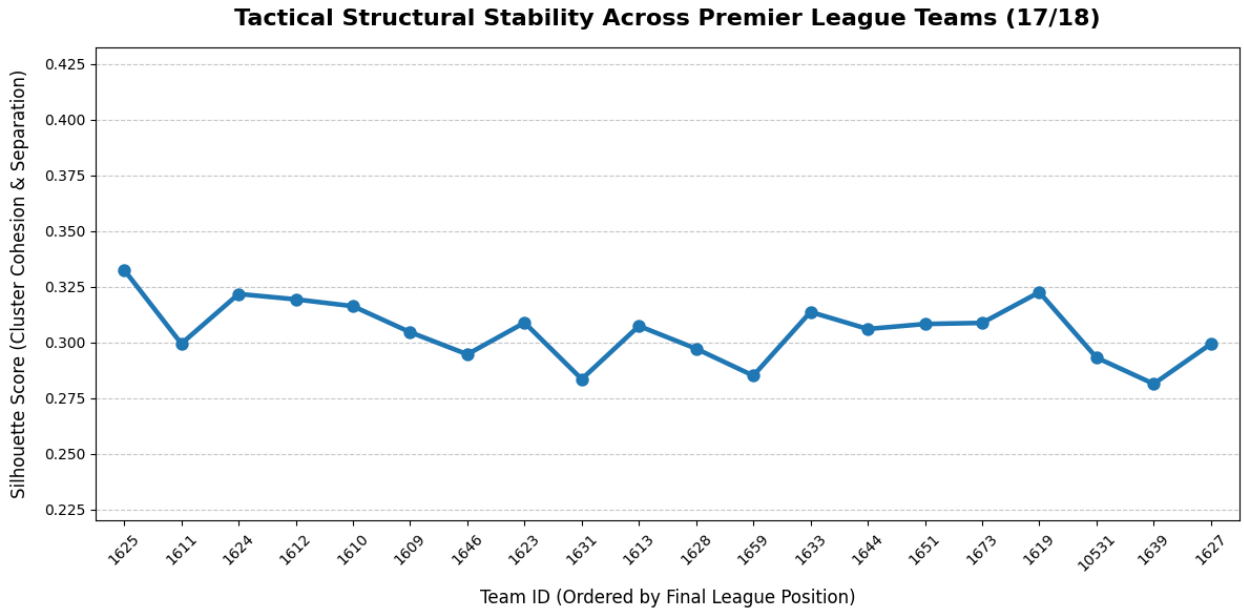


Figure 3: Silhouette score across all 20 teams

5.2 Network Dynamics & Transition Flow

To extract differences in playing strategies, we analysed directed networks of two tactically opposite teams : Manchester City (League champions with possession based play) and Burnley(ranked 7th with direct long-ball play). Our following visualisations correctly distinguish their playstyle.

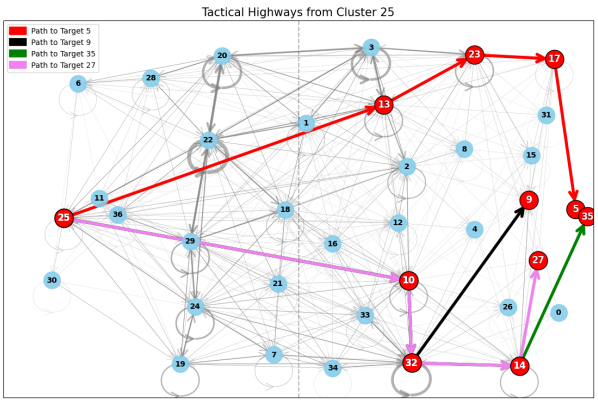


Figure 4: Burnley Flow Map

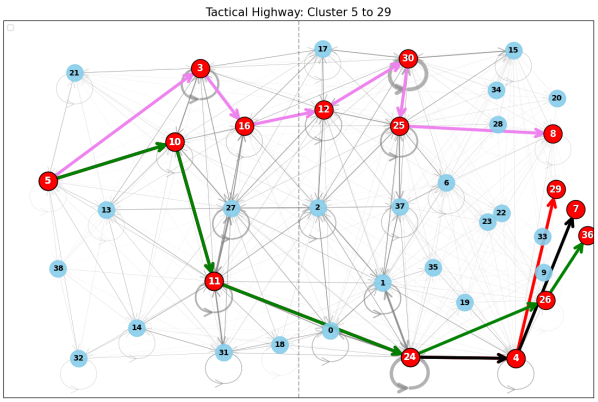


Figure 5: Man City Flow Map

Manchester City's graph shows a build up and possession based play. The shortest path from goalkeeper Ederson to the central striker's zone(Aguero) or second striker's zone(Jesus) flows through ball playing defender Otamendi, then distributed by Fernandinho to Kyle walker and then Walker carried it through right-flank and finally linked by Sterling or Bernardo Silva. On the other hand, Burnley started their attack from goalkepper Pope directly by Long Balls by surpassing midfielder's creative zones completely. Where Man City needed 6 steps on average to reach their attacking zone, Burnley used only 3-4 steps by playing long balls. This test validates our model's ability to identify playing patterns of distinct type of playing styles for different sides.

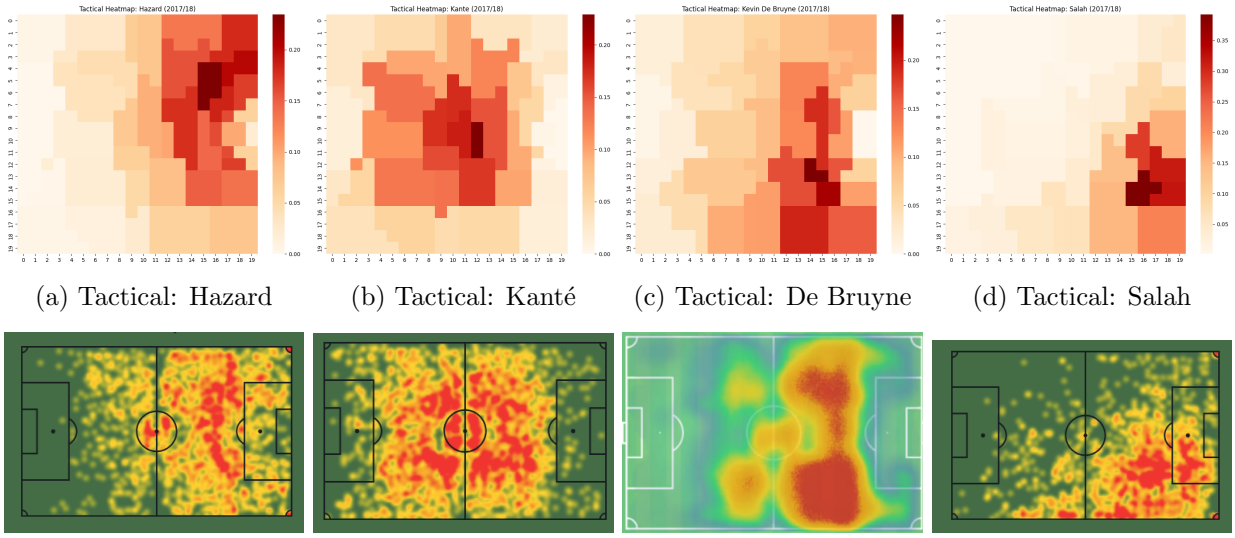
Table 3 presents the top 5 clusters with the highest Betweenness Centrality for both teams. For

Table 3: Comparison of Top 5 Clusters by Betweenness Centrality

Manchester City (Positional)		**Burnley (Direct)**	
Cluster	**Centrality**	**Cluster**	**Centrality**
24 (De Bruyne,Kyle Walker)	0.336	22 (Mee)	0.317
11 (Kompany)	0.296	32(Lowton)	0.261
4 (Kyle Walker)	0.238	24 (Long,Tarkowski)	0.209
25 (David Silva, De Bruyne)	0.194	3(Ward)	0.191
1 (Fernandinho)	0.186	23(Ward)	0.160

Manchester City highest betweenness scores are found in the right half-spaces and central midfield clusters. In 17/18 season Man City used their defenders as first line of attack. As a result Central backs like Otamendi completed 2500+ passes. Fernandinho ranked third in premiere league for total pass (2975 passes) with a success rate of 90.1%. In contrast, Burnley's central nodes are mainly captured by their central backs and side backs. Burnley's attacks starts with long ball from defenders like Mee, Tarkowski. Our model successfullly captures playing styles of both the teams very distinctively for both the teams.

5.3 Heatmap Test



(a) Tactical: Hazard (b) Tactical: Kanté (c) Tactical: De Bruyne (d) Tactical: Salah

(e) Real: Hazard (f) Real: Kanté (g) Real: De Bruyne (h) Real: Salah

Figure 6: Heatmap Recreation Test: Comparative analysis between model-generated tactical cluster occupancy (top row) and ground-truth heatmap (bottom row) for 2017/18 player archetypes.

By comparing heatmaps [Figure 6] of creators like Kevin De Bruyne and Eden Hazard with their original heatmaps we observed a high degree of similarity. For De Bruyne, he started most of their dangerous action from right flank with a very narrow link to the left flank. Our predicted heatmap captured this narrow link and right side dominance distinctively. On the other,hand, Hazard mostly operated on left half-space and outside of the box. Our heatmap also showed that thing by dark red clusters outside of the opponent box. Our model doesn't limit itself to only such creator's role only. For example, N'golo Kante's (Central defensive hub) tactical heatmap shows that he almost covered all blocks except Chelsea's box and opponent team's box. In 17/18 season Kante was one of those Central midfielders who engages on defensive actions as well as giving attacking threats. Our model even captures distribution of Inverted Winger Mohammed Salah. In a final note, all these results shows that our hierarchical clustering method captures team's playing styles as well as player's activity effectively.

6 Further Work

In future course of the study, we will use our clusters to integrate it with xT framework. We want to validate markovian dceision process framework and actual meaningful valuation of football actions to correctly evaluate expected threats from our clusters based on team playing tactics.

7 Code Availability

The code for this project is available at: here.

References

[1] S. Rudd (2011). A Framework for Tactical Analysis and Individual Offensive Production Assessment in Soccer Using Markov Chains. *New England Symposium on Statistics in Sports.*

[2] S. Green (2012). Assessing the Performance of Premier League Goalscorers. *www.statsperform.com.*

[3] G. Anzer and P. Bauer (2021). A goal scoring probability model based on synchronized positional and event data. *Scientific Reports*

[4] M. Scholtes and B. Karakuş (2024). Bayes-xG: player and position correction on expected goals using Bayesian hierarchical approach. *Frontiers in Sports and Active Living.*

[5] A. Rathke (2017). An examination of expected goals and shot efficiency in soccer. *Journal of Human Sport and Exercise.*

[6] K. Singh (2018). Introducing Expected Threat (xT). *Karun.in*

[7] L. Gyarmari and R. Stanojevic (2016). QPass: a Merit-based Evaluation of Soccer Passs.

[8] L. Pappalardo (2019). A public data set of spatio-temporal match events in soccer competitions. *Scientific Data*

[9] Markovian Decision Process. *Wikipedia*

[10] L. Hubert and P. Arabie (1985). Comparing partitions. *Journal of Classification*, vol. 2, no. 1, pp. 193–218.

[11] P. J. Rousseeuw(1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65.